Department of CSE(AIML), PES University
UE26CS151A -Python for Computational Problem Solving
Laboratory Week 6

Note: Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **marks**.

Problem Statements

Q1. Count the Number of Times Each Number Appears in a Text File Write a program to read integers from a text file and use a dictionary to count and display the frequency of each number.

- **User Input:** File name entered by the user.
- **Data Types:** Use dict with int keys (numbers) and int values (frequency).

Q2. Display Students Who Have Not Passed Using a Dictionary Write a program to store student names and marks in a dictionary and display the names of students who scored less than 40 marks.

- **User Input:** Student names (string) and marks (int) entered by the user.
- **Data Types:** Use dict with string keys and int values.

Q3. Find the Total Revenue from a CSV File Write a program to read product names, quantities sold, and prices from **week_6_set5_q3.csv** and calculate and display the total revenue from all products.

- **User Input:** No user input required.
- **Data Types:** Use string for product names, int for quantities, and float for prices.
- **File Reference:** Refer to **week_6_set5_q3.csv** for input.

Q4. Find the Most Frequent Word in a Text File Write a program to read a text file and use a dictionary to identify and display the word that occurs most frequently.

- **User Input:** File name entered by the user.
- **Data Types:** Use dict with string keys (words) and int values (frequency).

Q5. Display Students with Missing Marks from a CSV File Write a program to read student names and marks from **week_6_set5_q5.csv** and display the names of students whose marks field is empty.

- **User Input:** No user input required.
- **Data Types:** Use string for student names, int for marks (nullable).
- **File Reference:** Refer to **week_6_set5_q5.csv** for input.

Q6. Combine Inventory Dictionaries Write a program to create two dictionaries containing product names and their stock quantities. Combine the dictionaries by adding the quantities when a product appears in both dictionaries, and display the final inventory.

- **User Input:** Product names (string) and stock quantities (int) entered by the user.
- **Data Types:** Use dict with string keys and int values.

Q7. Separate Positive and Negative Numbers into Different Text Files Write a program to read integers from a text file and write the positive numbers into one text file and the negative numbers into another text file. Ignore zeros.

- **User Input:** Input file name (string) and output file names (string) entered by the user.
- **Data Types:** Use int for numbers.
- **File Handling:** Use text file read mode (r) and write mode (w).

Q8. Write a Python program that asks the user to enter the titles of three books and the number of pages in each. Store the data in a list of lists and write the results into a CSV file named week_6_set5_q8.csv.

- **Data Types: string for book titles, int for pages.**

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